

Part 6: Business Insights

1. Overall Survival

The number of passengers who did not survive was much higher than the number who survived. This shows that the overall survival rate on the Titanic was quite low.

2. Gender and Survival

Female passengers had a much higher survival rate than male passengers. This shows that gender played an important role in the chances of survival.

3. Passenger Class and Survival

Passengers from higher classes had better survival rates than passengers from lower classes. First-class passengers were more likely to survive, while third-class passengers had the lowest survival rate.

4. Fare and Survival

Passengers who paid higher fares generally had better chances of survival. This is also connected to passenger class because higher-class passengers usually paid more.

5. Age and Survival

Age also showed some differences between survivors and non-survivors. Children and younger passengers had different survival patterns compared with older passengers, showing that age was also related to survival.

6. Travelling Alone

Passengers travelling alone and passengers travelling with family had different survival rates. This suggests that travelling with family may have affected a passenger's chances of survival.

7. Embarkation Port

The survival rate was not the same for passengers from all embarkation ports. Some ports had a higher survival rate than others, which may be related to the type of passengers who boarded from each port.

8. Passenger Class and Fare

There was a clear relationship between passenger class and fare. First-class passengers generally paid much higher fares, while third-class passengers paid lower fares.

9. Family Size

Family size also showed different survival patterns. Passengers travelling with small families had different outcomes compared with people travelling alone or with larger families.

10. Relationship Between Numerical Variables

The correlation heatmap showed that some numerical variables were more closely related than others. In particular, passenger class and fare had a noticeable relationship, which makes sense because the class of a passenger was connected to the fare they paid.

Part 7: Business Recommendations

1. Improve Safety for Lower-Class Passengers

- Give passengers in lower-class areas clear and easy-to-follow emergency instructions.
- Make sure emergency exits and safety equipment are easily accessible in all passenger areas.
- Place enough staff in these areas to guide passengers quickly during an emergency.

2. Make Emergency Plans Family-Friendly

- Help families stay together during emergency situations.
- Give extra guidance to passengers travelling with young children.
- Clearly mark evacuation routes so families can reach safety without confusion.

3. Give Extra Support to Children and Older Passengers

- Identify areas where children and older passengers are likely to need more assistance.
- Train staff to provide additional help to passengers who may have difficulty evacuating quickly.
- Keep suitable evacuation routes available for passengers who need extra support.

4. Provide Equal Access to Safety Facilities

- Safety facilities should be easily accessible regardless of passenger class or fare paid.
- Place clear emergency signs throughout the ship.
- Make sure every passenger receives basic safety instructions before the journey starts.

5. Use Passenger Data for Emergency Planning

- Use information such as age, gender, passenger class, and family size to plan staff positions.
- Identify areas that may need more assistance during an emergency.
- Use passenger information to prepare better evacuation and rescue plans.